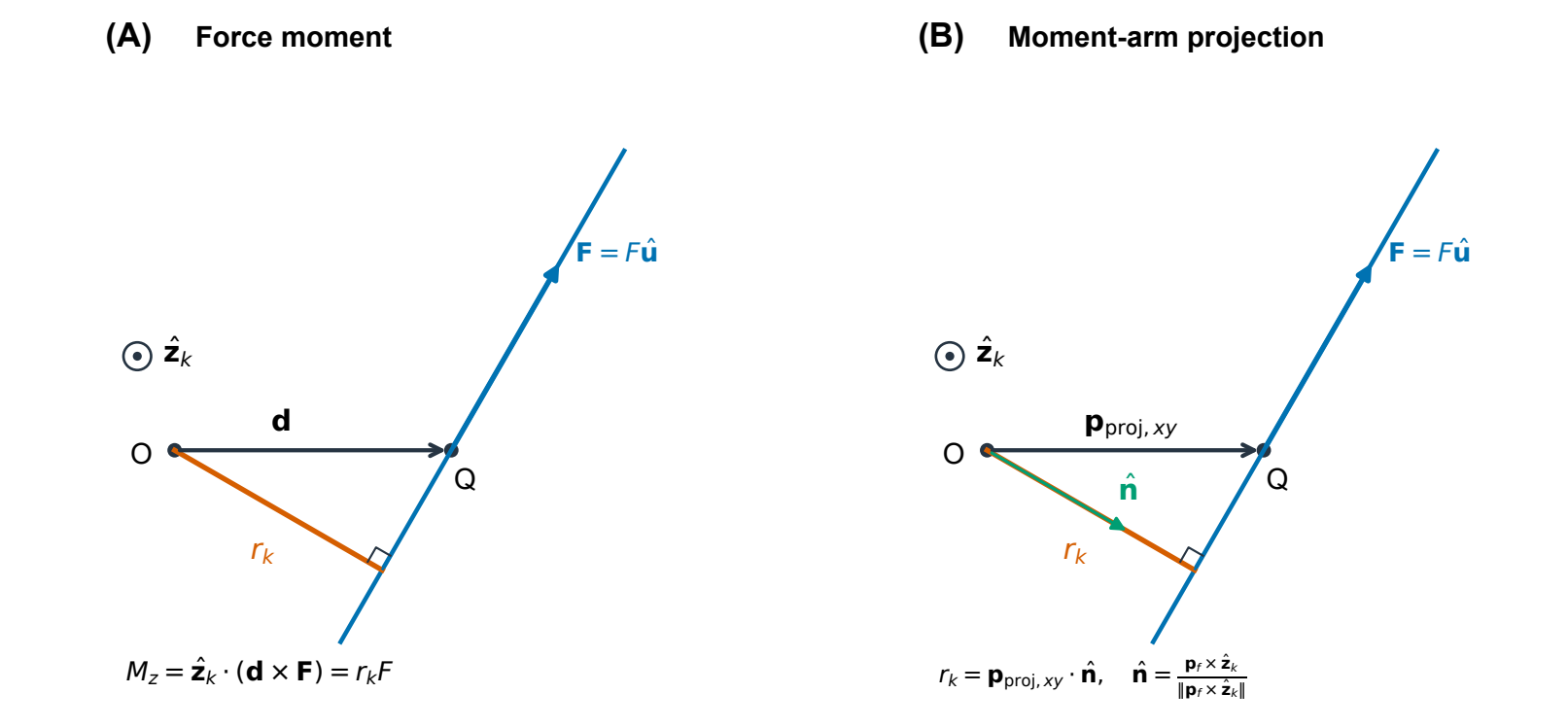
(A) Force moment
F = Fû
ẑk
d
O
Q
rk
Mz = ẑk · (d × F) = rkF
(B) Moment-arm projection
F = Fû
ẑk
pproj, xy
O
Q
n̂
rk
rk = pproj, xy · n̂, n̂ = pf × ẑk / ‖pf × ẑk‖